# KPlaceNet

## A geographic region from one photograph

Image-based geolocation with ResNet-50 and geographic cells

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OSV-5M photograph, CC-BY-SA-4.0. Project group, CSE 311.

# The prediction pipeline

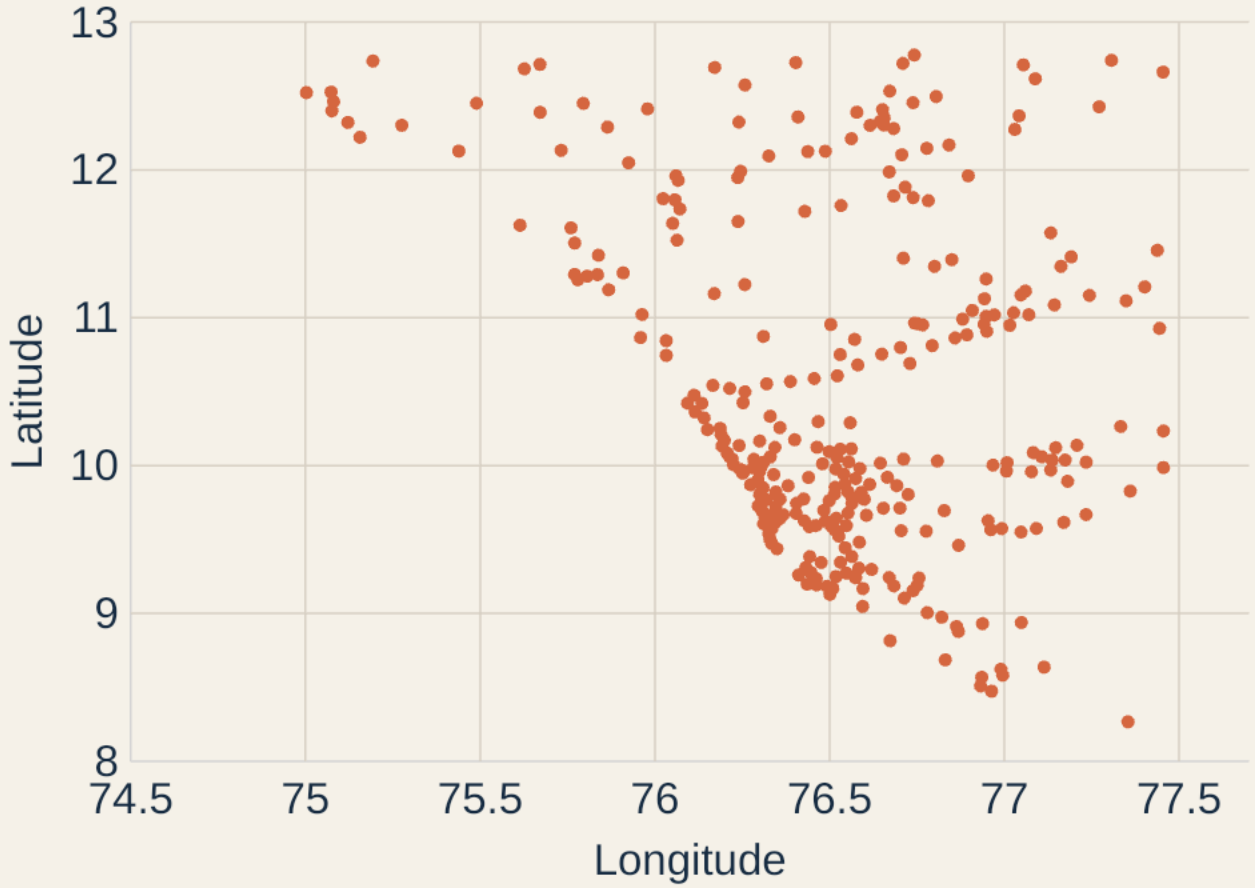
**RGB image** → **ResNet-50** → **2,048 features** → **300 cell scores** → **Cell centroid**

224 × 224 pixels

Coordinates supply training targets.
The network receives image pixels at inference.

Source: src/dataset.py, src/model.py and demo/app.py

# Geography becomes 300 classes



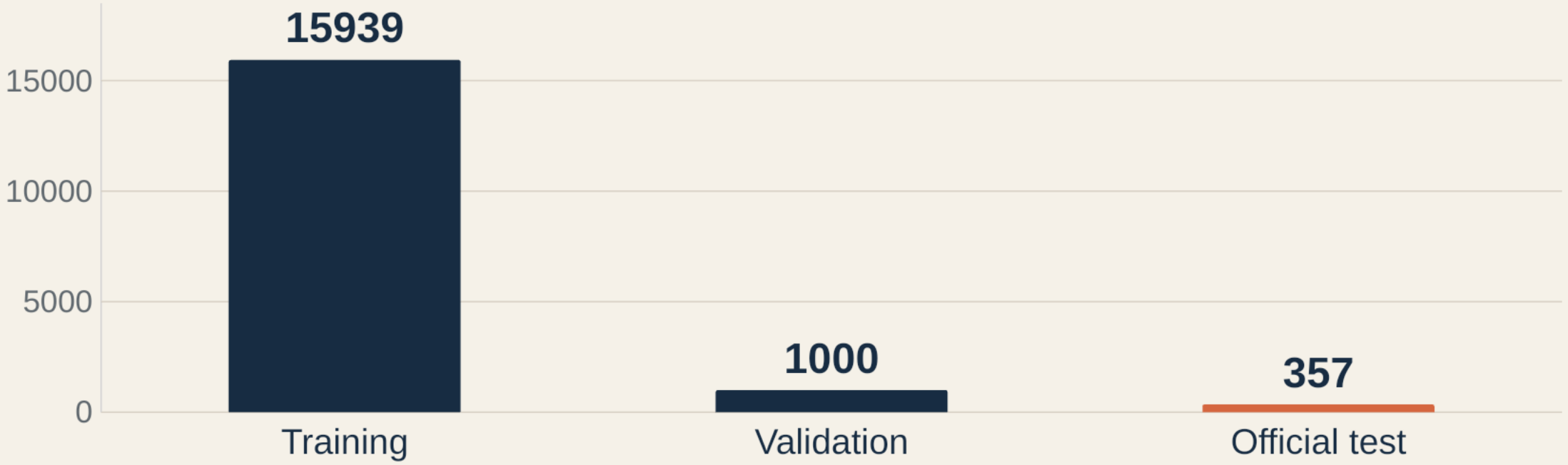

**Training coordinates form the class map**

The winning class returns its saved center.

The class resolution limits coordinate precision.

Source: released Kerala cells.json. Points show actual saved centroids.

# Kerala training, validation and test data




**Same images and same 300 cells for L5 and L6**

Source: reconstructed seed-42 CSVs and split_audit.json

# L6 gives nearby cells some training credit

## 90% assigned class

## +

## 10% geographic neighborhood

q = 0.9 one_hot + 0.1 softmax(-distance / 15 km)

Source: scripts/run_l6_kerala_head.py and prespecified protocol

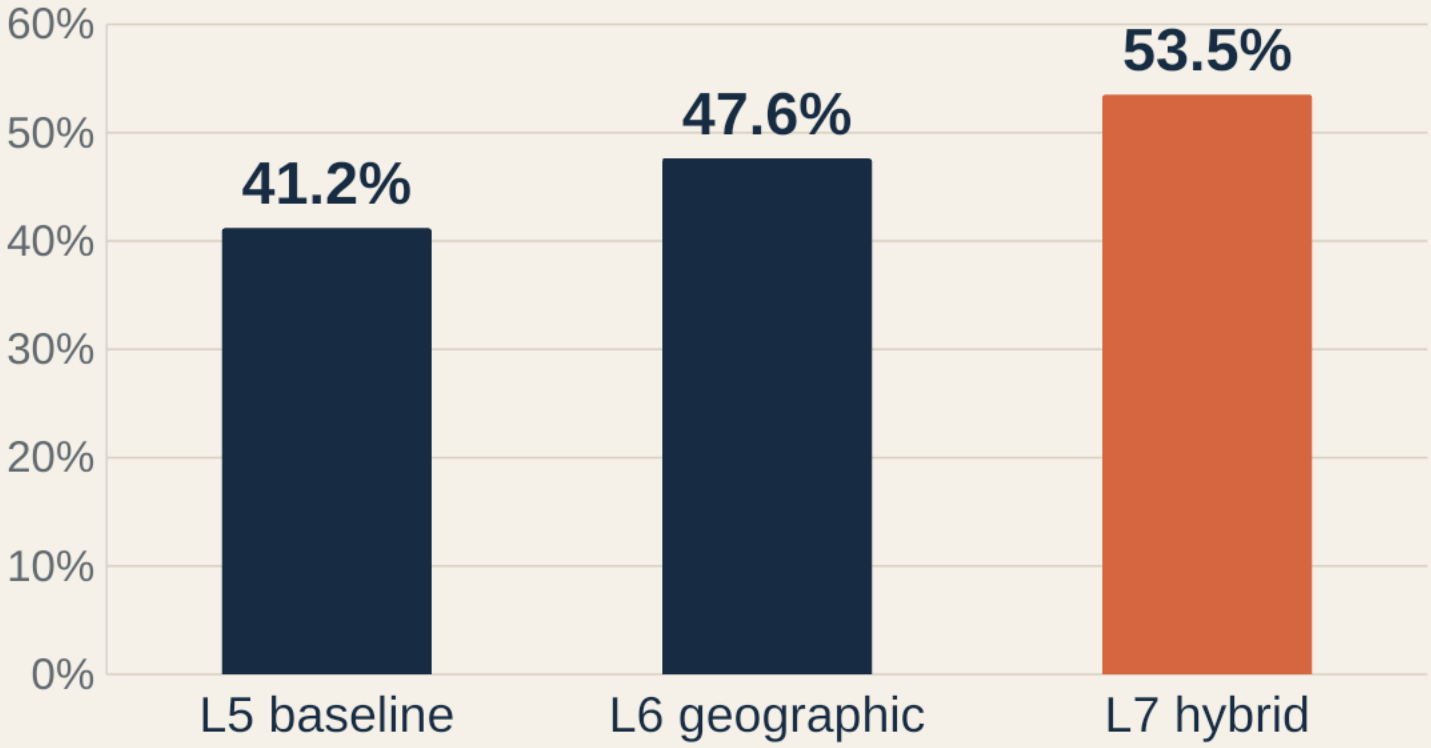
# L6 selection uses validation

**Fixed backbone** → **Two fixed recipes** → **Validation selection** → **Test after selection**

Selected: geographic smoothing, epoch 12

Source: protocol.json and selection_before_test.json

# More test photographs fall within 25 km



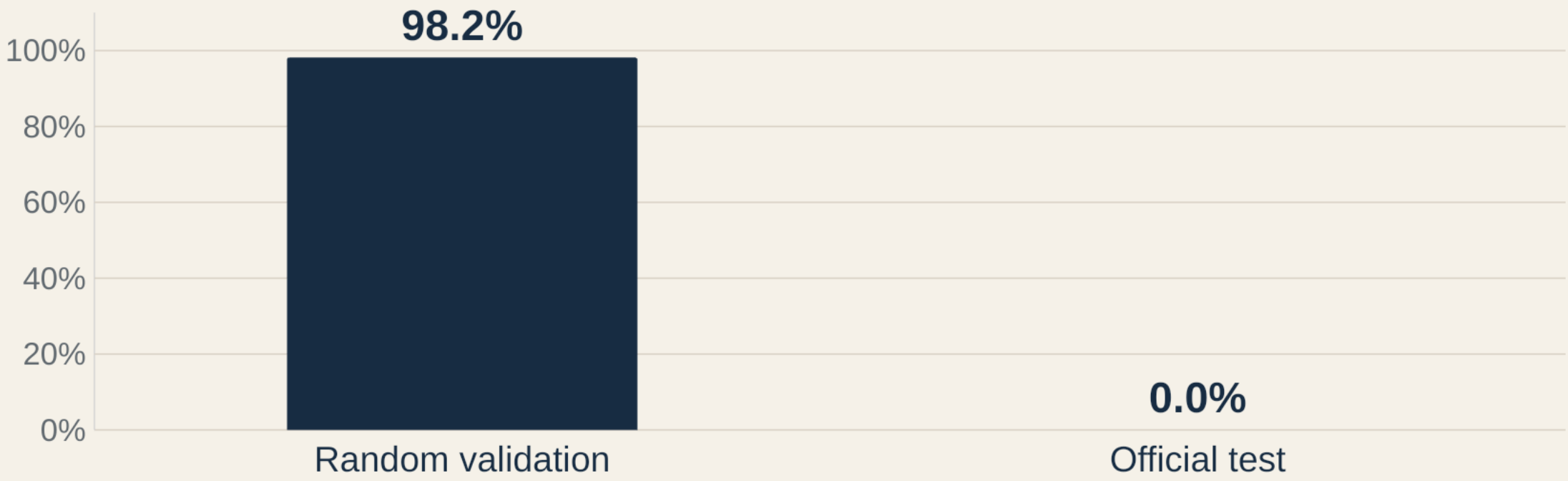

147 → 170 → 191 correct images out of 357

**21.8 km**

Hybrid median error
L6 alone: 28.4 km

Source: paired official test, 357 images. One seed.

# Random validation contains nearby street views




**Photos within 1 km of a training coordinate**

Source: nearest-training-coordinate audit of the recovered subset

# Confidence and distance measure different things

| Quantity | What it measures |
| --- | --- |
| Raw model score | Relative preference among cells |
| Within-25-km accuracy | Labeled test locations near enough |
| Conformal coverage | Truth appears in the candidate set |

## Kerala scores remain uncalibrated

Source: demo/app.py and the recorded global uncertainty study

# The ETH paper adds aerial imagery and retrieval

## KPlaceNet

Image features → Cell classifier → Cell center

## ETH research

Ground prototypes and aerial features → Fine-resolution retrieval

Source: Lindenberger et al., NeurIPS 2025. arXiv:2510.26795.

# The live demo displays a regional estimate



**Predicted cell 196**

**9.6761° N**
**76.4041° E**

Open the coordinate in Google Maps

OSV-5M image 1083931132408866, official test. CC-BY-SA-4.0.

# Better data is the next improvement

**350**

## geotagged candidate records

Coordinates, licenses and duplicates need review before the next training split.

Source: Wikimedia Commons candidate manifest and L6 limitations

# Training-photo neighbors refine the guess

## 80% classifier

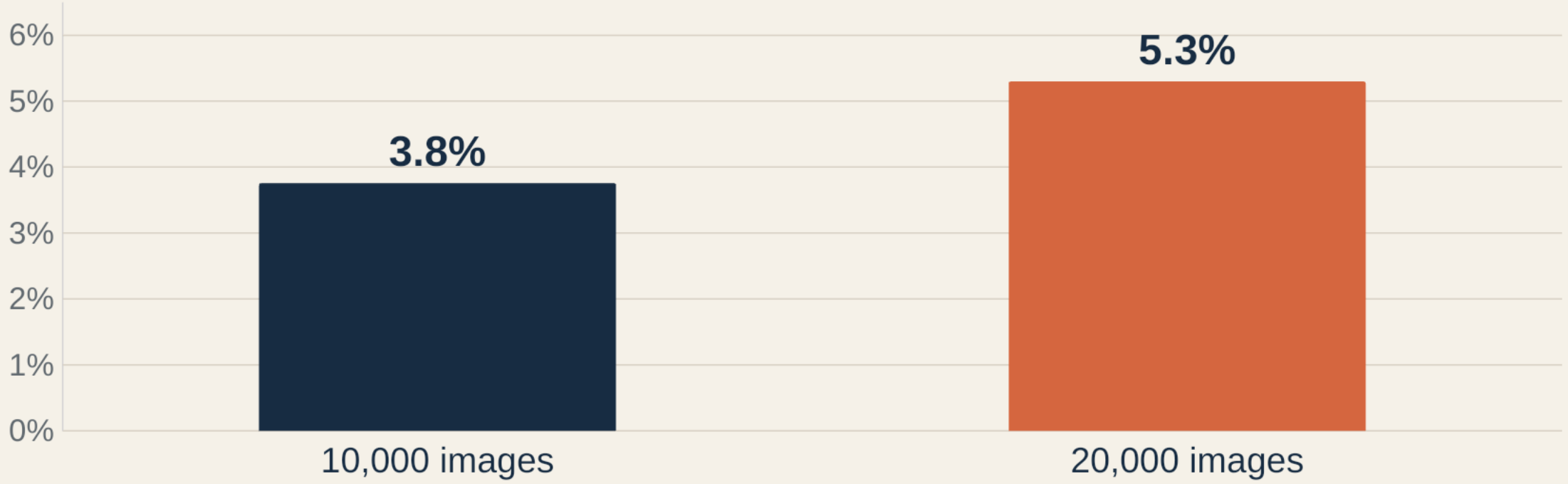
## + 20% neighbor votes

Five cosine-nearest ResNet features
from training photographs only

Retrieval improves this benchmark; it cannot verify the input region.

Source: fixed L7 protocol; training-only bank; reused official test benchmark.

# Appendix: the controlled global data-scale study




Within 200 km, averaged over three paired seeds

Source: three paired seeds, fixed 300 cells, recorded 9 October study

# Appendix: the finite-sample conformal cutoff

## k = ceil((n + 1)(1 - alpha))

Four scores: 0.1 0.2 0.4 0.8

alpha = 0.2 gives k = 4 and cutoff = 0.8

If k exceeds n, the conservative cutoff is infinity.

Source: src/uncertainty.py. Worked example, not model results.